Finding a Signal

Counting, cuts and significance — Lecture 16 (Chapter 8, Parts 2–3)

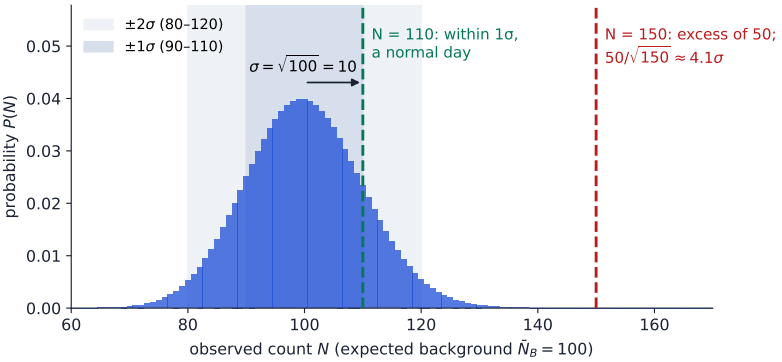
Amir Farbin

Part 2 — Counting, cuts and significance

Is there a second population?

- A dataset with **two populations**, one of them rare
  - an online store: **customers** (background) and **bots** (signal)
  - the LHC: the **Standard Model** (background) and **supersymmetry** (signal)
- The question: is the rare one there? How sure are we? How do we make the search as sensitive as possible?
- The key ingredient: an estimate of the **expected background** $\bar{N}_B$
  - the store: previous data · the physicist: physics calculations + simulation

Counting



- Expected $\bar{N}_B = 100$ a day; observed $\hat{N}$ fluctuates: Poisson, $\sigma = \sqrt{\bar{N}_B} = 10$
- 110 is a normal day. 150 is very unlikely to be a fluctuation

Significance

- Excess over the background: $\hat{N}_S = \hat{N} - \bar{N}_B$, uncertainty $\sigma = \sqrt{\hat{N}} \approx \sqrt{\bar{N}_S + \bar{N}_B}$
- **Significance**: how many standard deviations the excess is

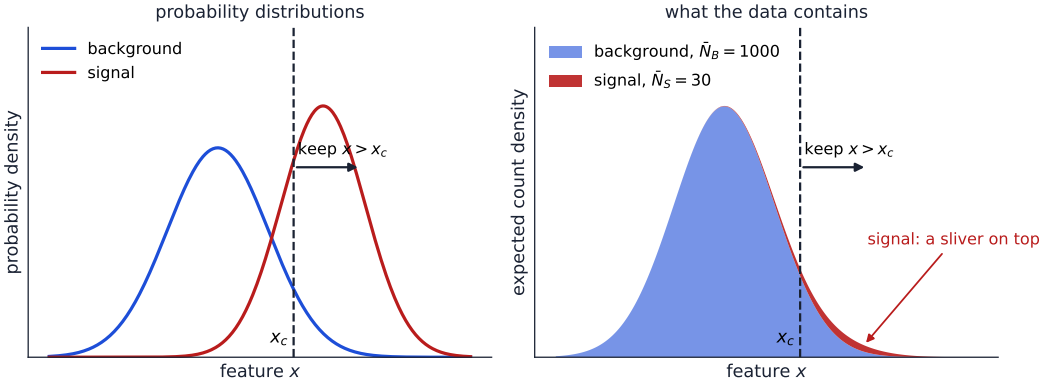
$$Z = \frac{\hat{N}_S}{\sqrt{\hat{N}}} \approx \frac{\bar{N}_S}{\sqrt{\bar{N}_S + \bar{N}_B}} \qquad \text{150 observed: } Z = \frac{50}{\sqrt{150}} \approx 4.1$$

- 3σ: **evidence**. 5σ: **observation** — a background fluctuation that large happens once in 3.5 million
- A convention from experience, not a law. The Higgs was announced at 5σ

Counting isn't enough

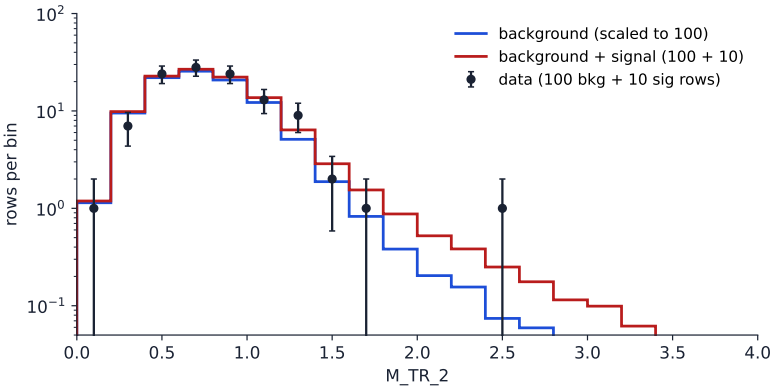
- The J/ψ was found (1974) by counting: an experiment's trigger rate suddenly jumped
- At the LHC: 40 million crossings a second, interesting things 10^{11} times rarer
  - the fluctuations of the total swamp the signal
- So **select**: if 900,000 of a million visitors are verified customers, look for bots among the other 100,000
- Every selection aims at the same thing: a better **signal-to-background ratio** in what's left

One feature, two pictures



- Left: the **probability distributions**. Right: what the **data** actually contains
- Cut too far right: very pure, but no signal left. Find the cut that **maximises significance**

What the experiment would see



- 10 signal + 100 background from the simulation, as a pretend experiment; error bars $\sqrt{N}$ per bin
- Two hypotheses: **background only** vs **background + signal**. Which does the data follow?

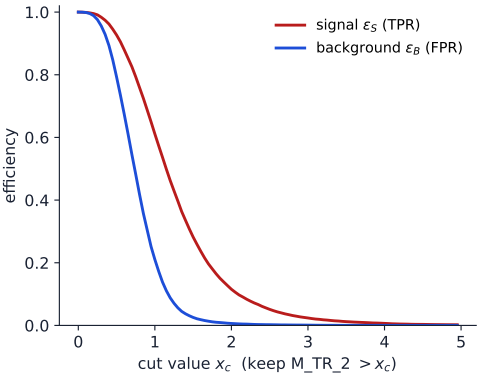
Efficiencies (TPR, FPR)

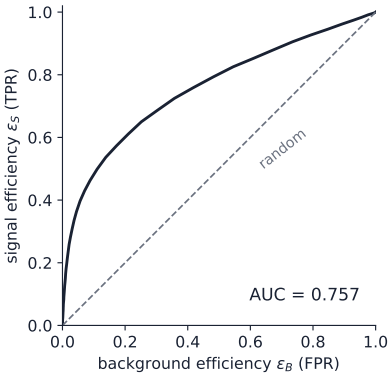
- For a cut $x > x_c$, the fraction of each population that survives:
  - **signal efficiency** $\epsilon_S(x_c)$ = true positive rate · **background efficiency** $\epsilon_B(x_c)$ = false positive rate
- After the cut, multiply: $\bar{N}'_S = \epsilon_S \bar{N}_S$, $\bar{N}'_B = \epsilon_B \bar{N}_B$

$$Z(x_c) = \frac{\epsilon_S \bar{N}_S}{\sqrt{\epsilon_S \bar{N}_S + \epsilon_B \bar{N}_B}}$$

- Only the efficiencies depend on x_c. Each is a **cumulative sum** of a histogram, from the right

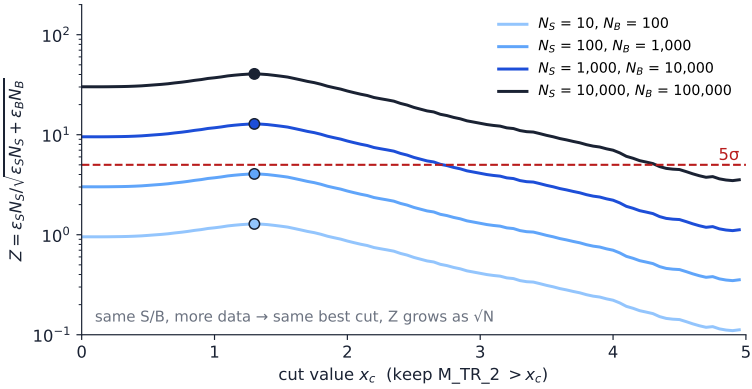
Efficiency curves and the ROC





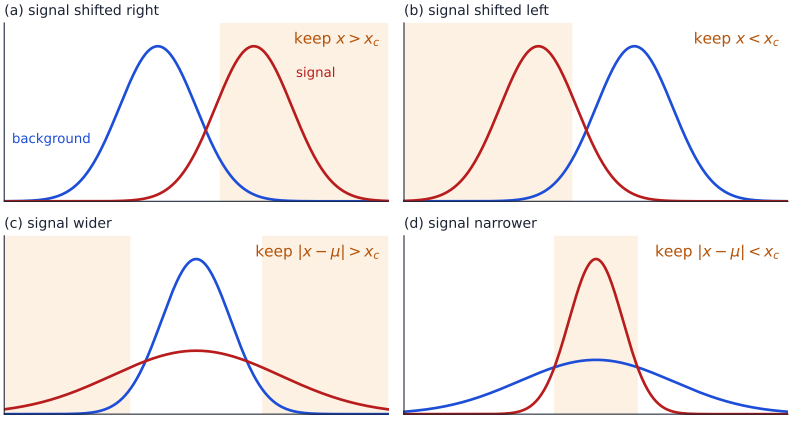
- Each point on the ROC curve is one choice of cut. **AUC**: 1 is perfect, 0.5 is a coin toss
- The same plot will judge every machine-learning classifier: its output is just another feature

Where to cut



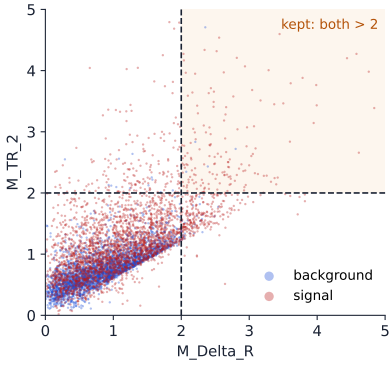
- Same signal-to-background ratio $\rightarrow$ **same best cut**. More data $\rightarrow$ Z grows as $\sqrt{N}$

Which cut?



- Look at the distributions first — that's what the pair plots were for

More than one feature



	M_TR_2 > 2	M_Delta_R > 2	product	both
signal	0.115	0.133	0.0153	0.0506
background	0.00626	0.0143	8.95e-05	0.00248

efficiency = fraction of rows passing;
'product' assumes the two cuts are independent

- Independent features: the fraction passing both = the **product**. It's about **3× larger for signal, 20–30× for background**
- Correlated features can't be cut one at a time; and each cut costs signal

Recap of Part 2

- Counting: excess over the expected background, in units of $\sqrt{N}$: the significance
- A cut on a feature improves signal-to-background; efficiencies are TPR and FPR
- Choose the cut that maximises Z; the ROC curve and AUC compare features
- Correlated features don't combine by multiplication

Part 3: from many features to one number — test statistics

Part 3 — Toward machine learning

Test statistics

- Every feature we cut on is a **test statistic**: a random variable used to perform a test (*is it signal?*)
- Many came from years of **feature engineering**; many are correlated
- Wanted: **one** number from all the features, $t = f(x_1, \ldots, x_n)$, better than any of them
  - then cut on t exactly as on `M_TR_2`
- f written by hand: feature engineering. f **learned from data**: machine learning

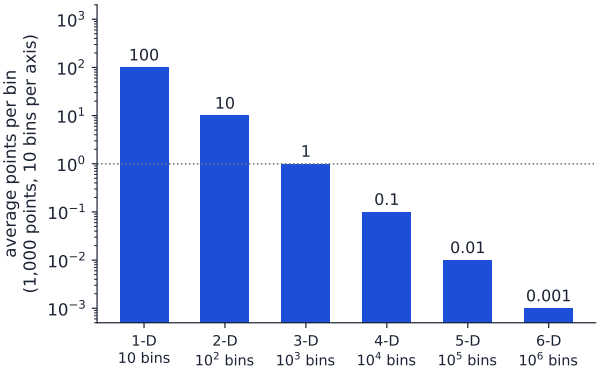
The best possible test statistic

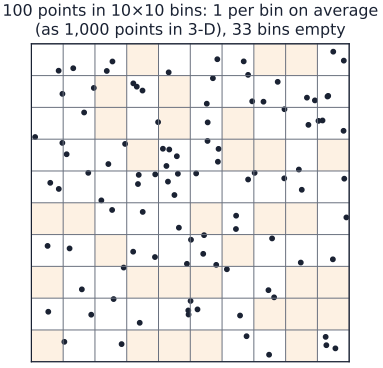
- If we knew the probability distribution of all the features under each hypothesis:

$$t = \frac{P(x_1, \ldots, x_n \mid \text{signal})}{P(x_1, \ldots, x_n \mid \text{background})} > t_c$$

- **Neyman–Pearson lemma**: nothing does better than this ratio
- But P is rarely known analytically: estimate it from labelled data or simulation — a histogram

The curse of dimensionality





- 10 bins per feature: 10^d bins in d dimensions. 1,000 events: 100 per bin in 1-D, 1 in 3-D
- 18 features: 10^{18} bins. No amount of simulation fills that

Two ways out

- **Feature engineering**: use domain knowledge to combine many features into one or two
  - the 10 high-level SUSY features
- **Machine learning**: learn the combination f from labelled data
  - the output is a test statistic: cut on it, draw its ROC, compute its significance
- Lab 7 asks for one by hand: **Fisher's linear discriminant** (Lecture 17)

Recap of Part 3

- A test statistic turns many features into one number to cut on
- The likelihood ratio is the best one, but needs probability distributions we can't estimate in many dimensions
- Feature engineering and machine learning both reduce dimensions; machine learning learns how

Next: machine learning (Chapter 9)